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A MULTI-ECHELON INVENTORY MODEL FOR A REPAIRABLE ITEM WITH ONE-FOR-ONE REPLENISHMENT*

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The problem that we address is to determine the inventory stockage levels in a multi-echelon inventory system for a repairable item. In its simplest form the multi-echelon system consists of a set of operating sites supported by a centrally-located repair depot. Each operating site requires a set of working items and maintains an inventory of spare items. The repair depot also holds an inventory of spare items. Item failures are infrequent and are replaced on a one-for-one basis. In this paper we present an exact model for finding the steady-state distribution of the net inventory level at each site. This model assumes that the failures are generated by a compound Poisson process and that the shipment time from the repair depot to each site is deterministic. No assumptions are made with regard to the repair cycle at the depot. We contrast this model with existing models for these systems. Based on the exact model we present an approximation for the steady-state distribution for the case with ample servers at the repair depot. We show that this approximation is very accurate on a set of test problems.

(MULTI-ECHELON INVENTORY SYSTEMS; REPAIRABLE ITEMS)

1. Introduction

In many industries and service organizations the reliance on multi-echelon inventory systems for repairing and supplying recoverable items is becoming more and more prevalent. The military's use of and interest in these systems is well known (e.g. Demmy and Presutti 1981, Clark 1981). Manufacturers of "high-technology" products also resort to multi-echelon inventory systems to support their field service operations. For instance, most computer and office equipment manufacturers maintain a service organization to service and repair their products in the field. Some of the service work done by these organizations requires the replacement of failed electronic modules that are repairable. Consequently, they have a multi-echelon inventory system consisting of field inventories for spare modules, and a centrally-located repair depot that repairs the modules and replenishes the field inventories. Communication networks also require multi-echelon inventory systems to ensure reliable service. For instance, a telephone system consists of a linked network of spatially-dispersed switching centers. Each switching center may have thousands of electronic modules that are each subject to failure. To maintain system reliability, the telephone system must stock some spare modules at every switching center. Yet for reasons of economies of scale, the system will also have a centralized repair facility and possibly intermediate buffers of spare inventory.

A key component in the design of multi-echelon inventory systems for recoverable items is the determination of the proper stockage levels of spare inventory at each echelon. A common approach to this problem has two components (Nahmias 1981). The first component is to characterize the service performance (e.g. expected shortages) of the multi-echelon system for a given specification of the inventory stockage levels. This may be done with an exact or approximate steady-state model. The second

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component is to search systematically over the possible inventory stockage levels to find the best choice with regard to both inventory costs and service performance.

In this paper we focus on the first component, namely the characterization of system performance for a given specification of the inventory stockage levels. In the next section we present a general framework for determining the steady-state distribution of net inventory levels, and relate this framework to the model developed by Simon (1971) and the METRIC model (Sherbrooke 1968). In §3 we discuss the computational implications of the general framework, in particular with regard to finding the best stockage levels. We also give an approximation to the steady-state distribution for a particular problem instance. We compare on a set of test problems this approximation with the approximation used by METRIC, and with the exact distribution. Our approximation seems very close to the exact distribution, and seems to dominate the METRIC approximation on the set of test problems. In the final section we discuss extensions to the general framework and directions for future research.

2. A General Framework

Consider a two-echelon inventory system for a repairable item where the system consists of a repair depot and N operating sites. Each site requires a set of working items and maintains an inventory of spare items. All failed items are repaired at the repair depot, which also maintains an inventory of spare items. We consider a one-for-one replenishment policy, which is appropriate when the item has high value and infrequent failures. When an item fails at a site, three events occur simultaneously. First, the failed item is replaced with a spare item from the site's inventory, if one is available; otherwise, there is a shortage at the site that will last until a replacement can arrive from the repair depot. Second, the failed item is sent to the depot for repair. Third, the depot ships a replacement item if it has available inventory; otherwise, the depot backorders the replacement request and will fill it when stock is available. When the failed item arrives at the repair depot, it enters the repair process; upon completion of the repair process, the item goes into the depot inventory or fills a backorder if any exist.

For this setting we require two critical assumptions. First we assume that at each site the failure process is a compound Poisson process that may depend upon the required number of working items, but does not depend on the actual number of working items. This assumption is typically violated whenever there are shortages at the site, i.e. the number of working items drops below the normal requirements. Nevertheless, this assumption is common to the literature (e.g. Sherbrooke 1968, Simon 1971, Muckstadt 1973, Allen and D'Esopo 1968, Richards 1976), and seems reasonable when the expected number of shortages at a site is small relative to the required number of working items at that site.

The second assumption is that the shipment time from the repair depot to each site is deterministic. Simon (1971) also makes this assumption, whereas the METRIC models do not (Sherbrooke 1968). Although we may permit this shipment time to vary across sites, for convenience we assume that the depot-to-site shipment time is the same for all sites and given by T_1.

We define $Q_i(t)$ to be the outstanding orders at site i at time t, that is the replacement requests that have yet to be filled. An unfilled request could be either in-transit from the depot to the site or backordered at the depot. If s_i is the number of spare items stocked at the site, then $s_i - Q_i(t)$ is the net spare inventory on hand at time t, where a negative value denotes a shortage level. Due to the first assumption, $Q_i(t)$ does not depend on the stockage quantity at the site s_i, but does depend on the number of spares stocked at the depot, denoted by s_0. For notational convenience we will make this dependence explicit only when needed.

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The primary result is

$$Q(t+T_1) \triangleq \sum_{i=1}^{N} Q_i(t+T_1) = B(t \mid s_0) + D(t, t+T_1) \tag{1}$$

where $Q(t+T_1)$ is the aggregate outstanding orders at the sites, $B(t \mid s_0)$ is the backorders at time t at the depot assuming the depot stocks s_0 items, and $D(t, t+T_1)$ is the aggregate failures at all sites over the time interval $(t, t+T_1]$. Hence, if $D_i(t, t+T_1)$ is the number of failures at site i, then

$$D(t, t+T_1) = \sum_{i=1}^{N} D_i(t, t+T_1).$$

$B(t \mid s_0)$ and $D(t, t+T_1)$ are independent random variables since the depot backorders at time t depend only on failures that occur prior to time t and since the failure process at each site is a compound Poisson process that is independent of the status of the site, and hence has independent increments.

We provide here an intuitive explanation of (1). At time t the outstanding orders at the sites either are in-transit to the sites or are backordered at the depot. All items that were in-transit at time t will arrive at the sites by time $t+T_1$. However, none of the depot backorders at time t can arrive at the sites by time $t+T_1$, since the shipping time from depot to site is T_1. For the same reason, any failure in the time interval $(t, t+T_1]$ generates an outstanding order that cannot be filled by time $t+T_1$. Hence, at time $t+T_1$, the depot backorders from time t, $B(t \mid s_0)$, and the aggregate failures, $D(t, t+T_1)$, must be outstanding. Furthermore, since $B(t \mid s_0)$ and $D(t, t+T_1)$ are independent, they do not double-count any orders, but do sum to give the total outstanding orders at time $t+T_1$. We note that this explanation is very similar in spirit to that for finding the on-hand inventory in a continuous-review inventory system with a constant lead time (Hadley and Whitin 1963, pp. 181–188).

This remarkably simple result requires only the assumptions of an independent compound Poisson failure process and deterministic shipment times from the depot to the sites. (Actually we can relax the first assumption to a failure process with independent increments.) We made no assumptions on the shipment times from the sites to the depot, or on the repair process. By using the same logic, we can extend the results to distribution networks with any number of echelons provided that the transit times from each echelon to its immediate successors are deterministic.

The utility of (1) for characterizing $Q_i(t)$ depends primarily on the resolution of two issues. First, we need the distribution for depot backorders $B(t \mid s_0)$ to convolve with the distribution for $D(t, t+T_1)$ (assumed to be known) to obtain the distribution of $Q(t)$. Second, we need to disaggregate $Q(t)$ into $Q_i(t)$ for $i = 1, 2, \ldots, N$. Knowledge of the distribution of $Q_i(t)$ permits us to find, for a given depot stockage level s_0, the best site stockage level s_i that optimizes some criterion. Before exploring these issues we first review the models of Simon (1971) and Sherbrooke (1968) in the light of (1).

Simon (1971) considers a two-echelon system with a Poisson failure process at each site, deterministic transit times, and ample repair capacity (i.e. there is no queueing, and successive repair times are i.i.d. random variables). He derives the steady-state distribution for the net inventory level at each site. In terms of (1), the depot backorders are

$$B(t \mid s_0) = [\, Q_0(t) - s_0 \,]^+ \tag{2}$$

where $Q_0(t)$ is the number of failed items in the system at time t and $[x]^+$ denotes the nonnegative part of x. $Q_0(t)$ consists of the (failed) items that are either in-transit from the sites to the depot or in the repair process at the depot. Whenever an item fails, the site requests a replacement from the depot. The depot then has an inventory outage

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from the time of the failure (or request) until the failed item is repaired. The net inventory level at the depot is just the difference between the planned stockage level and the number of failed items, i.e. $s_0 - Q_0(t)$. Hence, backorders occur once the failed items exceed the stockage level.

By assuming ample repair capacity, $Q_0(t)$ is equivalent to the occupancy level in an $M/G/\infty$ queue where the service time G includes both the in-transit time to the depot and the repair time at the depot. Since the steady-state distribution of $Q_0(t)$ is Poisson (Palm's theorem 1938), we find the steady-state distribution of $B(t \mid s_0)$ from (2). The convolution of $B(t \mid s_0)$ with $D(t, t + T_1)$ gives the distribution of aggregate outstanding orders $Q(t)$.

Simon assumes that the depot backorders are filled on a first-come, first-serve basis. This implies that the "disaggregation" of $Q(t)$ is equivalent to a *random* disaggregation across the sites. That is, the likelihood that any outstanding order is from a particular site i is directly proportional to that site's failure rate λ_i. To determine the distribution of $Q_i(t)$, we condition on $Q(t)$ and use the fact that the conditional distribution of $Q_i(t)$ is a binomial distribution. Specifically, we obtain

$$\begin{aligned}\Pr[\,Q_i(t) = j] &= \sum_{k=j}^{\infty} \Pr[\,Q(t) = k]\Pr[\,Q_i(t) = j \mid Q(t) = k] \\ &= \sum_{k=j}^{\infty} \Pr[\,Q(t) = k]\binom{k}{j}\left[\frac{\lambda_i}{\lambda}\right]^j\left[\frac{\lambda - \lambda_i}{\lambda}\right]^{k-j} \qquad (3)\end{aligned}$$

where $\lambda = \sum_{i=1}^{N}\lambda_i$ is the aggregate failure rate. Simon's original derivation also permits repair at the site and item condemnation (both features can be included in (1), but tend to obscure the simplicity of the result).

Kruse (1979) provides a similar interpretation of Simon's model and extends the model to more than two echelons. Shanker (1981) shows that Simon's model can permit compound Poisson demand, which is clear from this interpretation and the generalization of Palm's theorem by Feeney and Sherbrooke (1966).

Sherbrooke's METRIC model (1968) provides an approximate distribution for net inventory at each site for a two-echelon system with compound-Poisson failure processes and ample repair capacity. In that METRIC permits general transit times, it does not fit exactly into the framework given by (1). However, we can still review the METRIC model in terms of (1). For ease of presentation we suppose that we have Poisson failure processes. Then METRIC approximates the outstanding orders at each site, $Q_i(t)$, as the occupancy level of an independent $M/G/\infty$ queue; by Palm's theorem Q_i is Poisson. (This is an approximation since the times to fill outstanding orders are not independent.) From (1) we see that this implies that $B(t \mid s_0)$ is Poisson since both the failure process $D(t, t + T_1)$ and $Q(t + T_1)$ are assumed Poisson. Hence, the METRIC approximation for the case of a deterministic transit time to the sites is equivalent to approximating the depot backorder level by a Poisson random variable. The case with compound-Poisson failures is analogous.

3. Computational Issues

In the previous section we introduce a general framework for characterizing the performance of a two-echelon inventory system for a repairable item. In this section we consider the use of this framework to address the standard design questions for these inventory systems: how many spare items are needed and where should they be stocked? We explore both the use of exact and approximate models suggested by the framework, as well as the computational implications of these models.

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Exact Models

To use (1), we need a means for determining the backorder level at the depot $B(t \mid s_0)$, and a means for disaggregating $Q(t)$ into $Q_i(t)$. We discuss these issues here. Since we focus on steady-state distributions, we will drop the functional dependence on time.

The equation (2) defines backorders at the depot in terms of the failed items and the stockage level s_0. We note that the failed items *do not depend* on the stockage level s_0. Rather they depend only on the sites' failure processes (which are independent of the site status), and the repair cycle where the repair cycle consists of the shipping time from the site to depot, queueing time at the depot and the repair time at the depot. This independence property implies that we need determine the steady-state probability distribution of the failed items only *once* to obtain the steady-state probability distributions of the depot backorder level for all values of s_0.

Hence the application of (1) for any value of s_0 requires just the determination of the probability distribution of Q_0. Both Simon (1971) and Sherbrooke (1968) use Palm's theorem to characterize Q_0 under the assumption of ample repair capacity and Poisson or compound Poisson failure processes. For several other common scenarios, we can also determine the probability distribution of the number of failed items. For instance, suppose the sites' failure processes are Poisson, the shipment time from each site to the depot is a general random variable with finite mean, and there are k parallel repair lines, each with its repair time being exponentially distributed. A failed item arriving at the depot queues until a repair line is available. The distribution of the level of failed items is the convolution of the distribution of the number of items in-transit to the depot with the distribution of the number of items either in repair at the depot or in the repair queue. The number of items in-transit is the occupancy level in an $M/G/\infty$ system and hence has a Poisson distribution. Since the output of an $M/G/\infty$ is a Poisson process that is independent of the occupancy level (Mirasol 1963), we model the number of failed items either in queue or in repair as an $M/M/k$ system that is independent (in steady-state) of the number of items in-transit. The steady-state analysis of an $M/M/k$ system is straightforward. The convolution of the two distributions is trivial to do with a computer. Furthermore, this convolution need be done only once for the analysis of all stockage levels for a particular scenario. Tractable extensions to this problem setting include Erlang repair times and more complex repair processes that require a series of repair steps at distinct facilities.

Once we have found the distribution of the aggregate outstanding orders via (2) and (1), we need to disaggregate this distribution into the distributions of outstanding orders for individual sites. A general form for this disaggregation is

$$\Pr(Q_i = j) = \sum_{k=i}^{\infty} \Pr(Q = k)\Pr(Q_i = j \mid Q = k)$$

where we need specify the conditional distribution $\Pr(Q_i = j \mid Q = k)$ to reflect the depot priority scheme. If the depot fills replacement requests on a first-come, first-serve basis, then we use (3) where the conditional distribution is a binomial distribution. (Indeed (3) applies to any priority scheme that does not depend upon which site a resupply request came from.) For other priority schemes, such as filling first the most "urgent" request, the form of the conditional distributions is less clear, but presumably more complex.

The computation required to obtain these distributions is not trivial and must be done for every value of s_0 that is investigated. To ease this computational burden we present next an approximation to the disaggregation process.

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Approximate Model

For the approximate model we assume that all failure processes are Poisson and that the depot fills requests on an FCFS basis so that (3) gives the disaggregation of the aggregate outstanding order process. From (3) and (1) we show in Graves (1983) that the expected level of outstanding orders at a site and its variance are given by

$$E\{Q_i\} = \frac{\lambda_i}{\lambda} E\{B(s_0)\} + \lambda_i T_1, \tag{4}$$

$$\mathrm{Var}\{Q_i\} = \left(\frac{\lambda_i}{\lambda}\right)^2 \mathrm{Var}\{B(s_0)\} + \left(\frac{\lambda_i}{\lambda}\right)\left(\frac{\lambda-\lambda_i}{\lambda}\right) E\{B(s_0)\} + \lambda_i T_1 \tag{5}$$

where λ_i is the failure rate for site i, $\lambda = \sum_{i=1}^{N} \lambda_i$ is the aggregate failure rate, and T_1 is the deterministic shipment time from depot to site. The notation $B(s_0)$ denotes the depot backorder level given s_0 units stocked at the depot. Hence we can express the mean and variance of each site's level of outstanding orders in terms of the mean and variance of the depot backorder level. Now from (2) we show in Graves (1983) that

$$E\{B(s_0)\} = E\{B(s_0-1)\} - \Pr(Q_0 \geq s_0), \tag{6}$$

$$\mathrm{Var}\{B(s_0)\} = \mathrm{Var}\{B(s_0-1)\} - [E\{B(s_0)\} + E\{B(s_0-1)\}] \cdot [1 - \Pr(Q_0 \geq s_0)]. \tag{7}$$

That is, we can compute the mean and variance of the depot backorder level recursively, provided we have knowledge of the distribution of the number of failed items. We propose to use (4)–(7) to compute the mean and variance of Q_i for each site for all values of s_0 of interest. We then intend to approximate the distribution of Q_i with a distribution that is fully specified by its first two moments. We illustrate this approximation strategy next.

We consider the case where all failure processes are Poisson and where there is ample repair capacity at the depot so that Q_0 has a Poisson distribution. An exploratory investigation of this case over a range of parameter sets suggests that the distribution of a site's level of outstanding orders Q_i will be unimodal and will have its variance strictly greater than its mean. We propose to approximate the distribution of Q_i by a negative binomial distribution; that is

$$\Pr(Q_i = j) = \binom{r+j-1}{j} p^r (1-p)^j \qquad \text{for} \quad j = 0, 1, 2, \ldots, \tag{8}$$

where r and p are positive parameters ($0 < p < 1$) such that

$$E\{Q_i\} = r(1-p)/p, \tag{9}$$

$$\mathrm{Var}\{Q_i\} = r(1-p)/p^2. \tag{10}$$

The mean and variance of Q_i are found from (4)–(7).

To test the effectiveness of this approximation we compare it with the METRIC approximation and with the exact distribution on a set of test problems. For all test problems, there are four sites ($N = 4$) and the shipment time T_1 from depot to each site is 3 days. The expected duration of the repair cycle (shipment time from site to depot plus repair time at depot) takes on one of four values: $E\{T_2\} = 1, 3, 6$, or 9 days. The aggregate failure rate is one of the four values: $\lambda = 0.5$, 1, 2, or 4 failures/day. The failure rates for the four sites are such that $\lambda_1/\lambda = 0.1$, $\lambda_2/\lambda = 0.2$, $\lambda_3/\lambda = 0.3$, and $\lambda_4/\lambda = 0.4$. To specify a test problem we need set the depot stockage level s_0. Since the aggregate failed items Q_0 is a Poisson random variable with a mean and variance equal

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to $\lambda E\{T_2\}$, we need set the depot stockage level to correspond to this level. To do this, we permit s_0 to range from $\lambda E\{T_2\} - (\lambda E\{T_2\})^{1/2}$ up to $\lambda E\{T_2\} + 2(\lambda E\{T_2\})^{1/2}$ (i.e. from $\mu - \sigma$ to $\mu + 2\sigma$). In particular, we let s_0 take on up to six integral values evenly-spaced over this range. Finally, for each test problem we specify the desired fill rate α for each site, where $\alpha = 0.84, 0.87, 0.90, 0.93, 0.96, 0.99$. Hence, for a given fill rate α the site stockage level s_i is the minimal quantity such that

$$\Pr\{Q_i < s_i\} > \alpha. \tag{11}$$

We chose to specify a desired fill rate rather than an expected-backorder level, because the fill rate is parameter-free while the expected-backorder level is not.

To recap, we specify a test problem by setting the expected repair cycle time (4 candidates), the aggregate demand rate (4 candidates), the depot stockage level (up to 6 candidates), and the site fill rate (6 candidates). This gives a maximum of 576 test problems. For each test problem, each analysis method (i.e., negative binomial approximation, METRIC approximation, and exact distribution) generates the required stockage level for each of the four sites. Hence, a maximum of $4 * 576 = 2304$ comparisons are possible.

The results from the test problems are summarized in Table 1. In total there are 1968 problem instances. We report the number of instances that each approximation yields the "wrong" stockage quantity for a site, that is, the approximation determines a stockage quantity that differs from that determined from the exact distribution. Overall both approximations are extremely effective. The METRIC approximation results in a wrong decision in 227 problem instances or 11.5% of the cases. In all of these instances the METRIC approximation recommends less stock than is actually required. The negative binomial approximation is even better. It errs in only 18 problem instances or 0.9% of the cases. Furthermore, the negative binomial approximation virtually dominates the METRIC approximation on these test problems. In all but two cases where the negative binomial approximation errs, the METRIC approximation also yields a wrong decision. In the two exceptional cases the negative binomial approximation recommends more stock than is actually needed. The computational requirements for the negative binomial approximation are comparable to that for the METRIC approximation. Whereas one requires both the mean and variance of each site's outage level [i.e. (4)–(5)], the other requires just the mean. The computation of the negative binomial distribution is of the same complexity as the computation of the Poisson distribution required by METRIC.

In conducting these computational tests we permitted the depot stockage level s_0 to be less than the expected number of failed items in the system, namely $\lambda E\{T_2\}$. For these cases, the safety stock at the depot, defined as $s_0 - \lambda E\{T_2\}$, is negative. Thus, on the surface, these test problems appear unreasonable since a stocking policy with negative safety stock may seem unthinkable. However, we found in our investigation that the stocking policy that satisfied the fill rate criterion with the minimal system inventory often had negative safety stock at the depot. In particular, for fill rates of less than 90%, the best stocking policy for the set of test problems almost always has a depot stockage level that implies negative safety stock.

Slay (1982, 1984) has independently proposed and studied the negative binomial approximation for this setting. This work predated the work reported here. His results concur with those given here, particularly with regard to the improvement of the negative binomial approximation over the METRIC approximation. Sherbrooke (1984) has extended this approximate model to the multi-indenture problem, and shows similar improvements over the MOD-METRIC approximation. In addition, other authors have noted the inaccuracy of the METRIC approximation, e.g., Kaplan (1980) and Shanker (1981).

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TABLE 1

Error Incidents for Approximation Methods

λ	$E\{T_2\}$	Number** or Problem Instances per Cell	Site Demand Rate			
			$\lambda_1 = 0.1\lambda$	$\lambda_2 = 0.2\lambda$	$\lambda_3 = 0.3\lambda$	$\lambda_4 = 0.4\lambda$
0.5	1	6	0,0*	0,0	0,0	0,0
	3	18	0,0	0,0	0,0	0,0
	6	36	0,0	2,0	0,0	2,0
	9	36	1,1	4,0	4,0	4,0
1.0	1	12	0,0	0,0	0,0	1,0
	3	36	3,1	1,0	4,0	4,1
	6	36	1,0	5,0	4,0	7,0
	9	36	3,0	5,0	6,1	12,2
2.0	1	24	0,0	0,0	0,0	3,1
	3	36	0,0	4,0	2,0	5,0
	6	36	2,0	1,1	10,0	10,1
	9	36	2,0	3,0	9,1	10,0
4.0	1	36	2,1	0,0	0,0	2,0
	3	36	2,0	3.1	5,0	8,1
	6	36	4,1	5,0	8,0	13,1
	9	36	4,1	6,0	13,0	18,2

* $\boxed{x, y}$: x = number of problem instances for which METRIC gave incorrect stockage quantity.

y = number of problem instances for which negative binomial approximation gave incorrect stockage quantity.

** For each choice of $E\{T_2\}$ and λ_i, we considered six fill rates and up to six values for the depot stockage quantity.

4. Discussion

The contribution of this paper is twofold. First, we provide a general framework for determining the distribution of net inventory levels in a multi-echelon system for recoverable items. Second, assuming ample servers at the repair facility, we show that the proposed approximate model is very accurate on a set of test problems. In this section we discuss extensions to the general framework and directions for future research.

The general model (1) requires a set of assumptions that may be restrictive. The assumption of a compound Poisson failure process is general enough to capture most failure processes. The assumption of a deterministic shipment time from depot to site is somewhat restrictive. Shipment times do vary but are reasonably predictable. Indeed, many manufacturers will ship small, high-valued, electronic modules via an air cargo service that guarantees overnight service. The assumption that the failure process is independent of the site status clearly does not hold if there is a small population of working items that generate the item failures. However, even in this case, if there is a very small probability that an operating site has a shortage, then it may be appropriate to model the failure process as being independent of site status. Finally, the assumption of a one-for-one replenishment policy does not allow emergency ordering. In

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many instances, we might consider more complex policies that permit emergency replenishments, but at a premium cost. Future work needs to examine the applicability of and extensions to the general model when its assumptions are not reasonable.

We presented the general model in its most simple context: two echelons, all failures are repairable, all repairs occur at the depot, and all depot-to-site shipment times are the same. The framework extends directly to M-echelon systems ($M > 2$) provided we have deterministic shipment times from each echelon to its successors. As an example, suppose we have three echelons with a single repair depot, multiple intermediate sites, and multiple operating sites. Then using (1) we determine the aggregate outstanding orders at the intermediate sites, where s_0 is the stockage level at the depot and T_1 is the shipment time from depot to the intermediate sites. We then disaggregate the aggregate orders at the intermediate sites, as in (3). Given a stockage level for each intermediate site, we can determine its backorder level. We now reapply (1) for each intermediate site and its successor operating sites to get the aggregate outstanding orders at the sites. Again we disaggregate for the sites, as in (3). Admittedly, the details are quite involved, but the general framework applies.

The general model can include site repair, and item condemnation and procurement. In the first case, we just augment the model (1) to include the random variable for the aggregate number of items in site repair. In the second case, the determination of the depot backorders must reflect the effect of condemned units. Both Simon and D'Esopo (1971) and Richards (1976) show how to find the depot backorder level when we include item condemnation.

If the shipment times are not the same, then we need restate (1) as $Q_i(t + T_i) = B_i(t \mid s_0) + D_i(t, t + T_i)$ where $B_i(t \mid s_0)$ represents the depot backorders at time t that are due at site i, and T_i is the shipment time from depot to site i. Hence, the outstanding orders at site i at time $t + T_i$ are those from time t that cannot arrive by time $t + T_i$, plus the failures over the interval $(t, t + T_i]$. We determine $B_i(t \mid s_0)$ for each site by disaggregating $B(t \mid s_0)$, as in (3).

We have suggested and tested one approximation for the two-echelon system with ample servers at the repair facility. We have not examined this approximation for cases with limited service channels or with more than two echelons. We hope that future research will address these cases.[1]

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